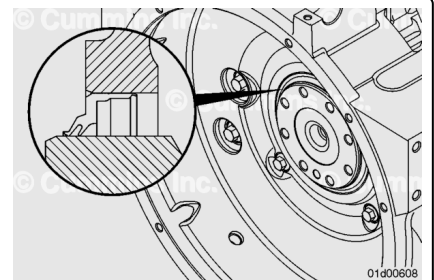


Trainings ▾

General Information

Lip Seal

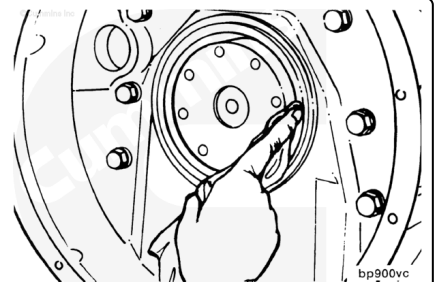
- Engines covered by this procedure use a lip style rear crankshaft seal (shown) in which the rotating portion of the sealing occurs at the contact surface between the lip of the seal and the crankshaft.
- The engine uses a rear crankshaft seal that is installed into the flywheel housing bore.



Some rust prevention oils applied at the factory have fluorescent tracer dyes added to them and can provide false test results when checking for external oil leaks using a black light.

Clean the area around the rear seal with solvent.

Check the area with a black light to insure there is **not** already fluorescent dye present before adding tracer dye to the lubricating oil.



Unitized Seal

Some engines used to use a unitized seal, which has a metal inner diameter that allows the inner and outer diameter of the seal to press-fit onto the crankshaft and the flywheel housing respectively, requiring service tool, Part Number 3164660, to remove.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Note : Use a container that can hold at least 26 liters [27 qt] of lubricating oil.

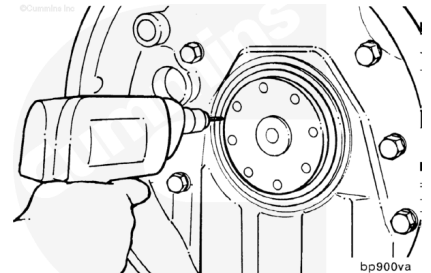
- Disconnect the batteries. See equipment manufacturer service information.
- If equipped with a wet flywheel housing, drain the oil from the flywheel housing by removing the plug in the bottom of the flywheel housing.
- Remove the transmission and all related components, if equipped. See equipment manufacturer service information.
- Remove the flywheel, if installed. Refer to Procedure 016-005 in Section 16.
(/qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html)
- Remove the flexplate, if installed. Refer to Procedure 016-004 in Section 16.
(/qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html)

Remove

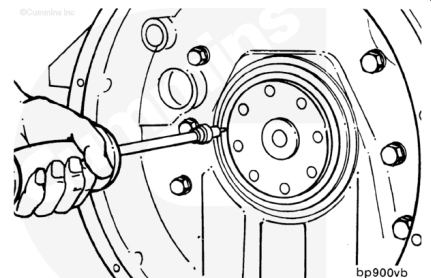
Lip Seal

Note : Lip style rear crankshaft seals can be removed using screws and a slide hammer. This procedure will not be followed on engines with a unitized seal.

Drill two holes, 180-degrees apart, into the rigid portion of the seal.



Install two coarse thread sheet metal screws into the holes just drilled. Use a slide hammer, Part Number 3823822, to remove the rear crankshaft seal.

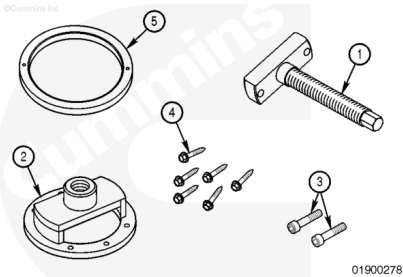


Unitized Seal

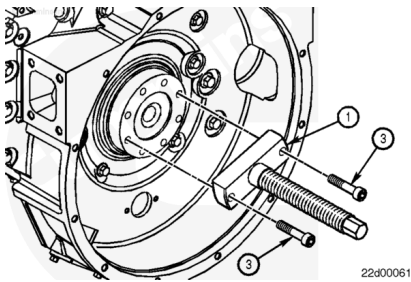
For engines with unitized seals, use service tool Part Number 3164660.

Table 2. Rear Crankshaft Seal Replacer Kit, Part Number 3164660

Item Number	Part Number	Description	Quantity
1	3164666	Replacer screw assembly	1
2	3164664	Crankshaft seal replacer	1
3	3164174	Socket head capscrew, M12 x 1.25x 25 mm	2
4	3164217	Sheet metal screw, Number 10x25.4 mm [1 in] long	25 (7 shown)
5	-	Disposable plastic driver (purchased with rear crankshaft seal kit)	1



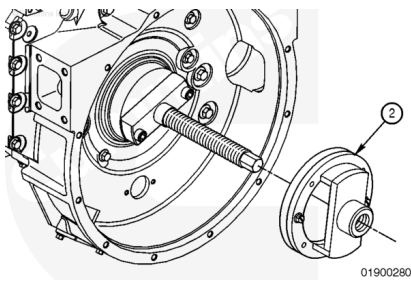
With the rear crankshaft seal installed onto the crankshaft flange as described earlier in this procedure, mount the replacer screw assembly (1) onto the rear of the crankshaft. Install the two M12 x 1.25 x 60-mm socket head capscrews (3).



⚠ CAUTION ⚠

Do not use an impact wrench or air tools. Doing so can damage the tool.

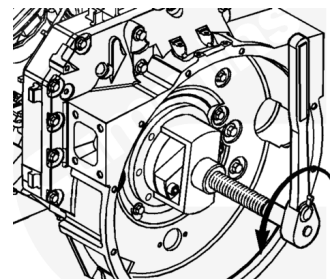
Lubricate the replacer screw with anti-seize compound or a suitable grease.



Hold the replacer screw and install the crankshaft seal replacer (2) onto the replacer screw assembly. Advance the crankshaft seal replacer toward the seal by rotating it **clockwise** until the attached disposable driver is positioned against the rear crankshaft seal. Then insert small screws into the crankshaft seal replacer tool.

⚠ CAUTION ⚠

Do not overtighten the replacer screw assembly after the crankshaft seal replacer contacts the flywheel housing. Doing so can damage the tool.



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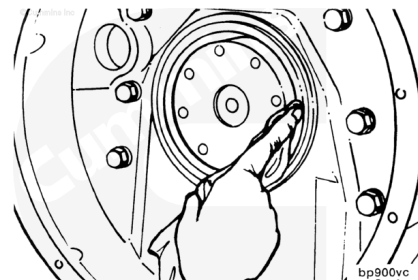
While holding the crankshaft seal replacer, rotate the replacer screw **counterclockwise** until the disposable driver attached to the crankshaft seal replacer makes contact with the flywheel housing.

Remove the service tools.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



bp900yc

⚠ WARNING ⚠

Compressed air used for cleaning should not exceed 207 kPa [30 psi]. Use only with protective clothing, goggles/shield, and gloves to reduce the possibility of personal injury.

⚠ CAUTION ⚠

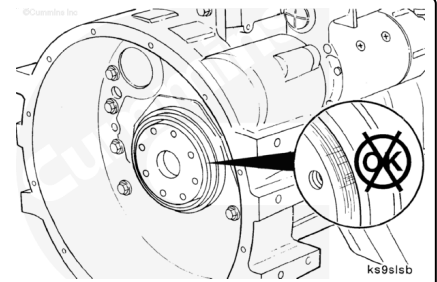
The seal lip and the sealing surface on the crankshaft must be free from all oil residue to prevent seal leaks.

Clean the rear crankshaft sealing surface and bore with solvent.

Dry the surface with compressed air.

Note : If the crankshaft has excessive wear, a service wear sleeve is available.

Inspect the crankshaft seal contact area for a wear groove. If the groove is deeper than 0.25 mm [0.010 in], a wear sleeve and oversize seal **must** be used. Refer to Procedure 001-067 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-067.html>)



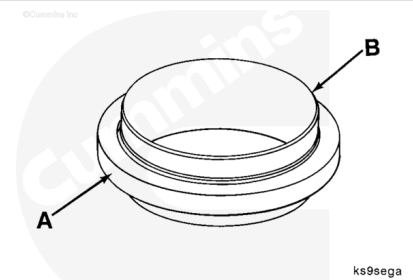
Install

⚠ CAUTION ⚠

The seal lip/bore and the sealing surface on the crankshaft must be free from all oil residue to prevent seal leaks.

To aid in installation, apply a mild soap (recommended 50/50 water to soap volumetric ratio) or P-80, Part Number 3824878, to the outside diameter of the seal case (A)

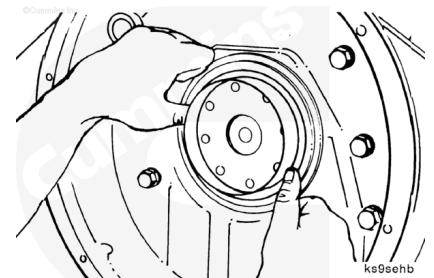
Note : A seal pilot (B) is provided with the new seal. The seal **must** be left on the seal pilot while installing the seal onto the nose of the crankshaft. This will make sure the lips of the seal do not get damaged during installation.



Place the new rear crankshaft seal, with the seal pilot, over the crankshaft nose and slide it by hand toward the flywheel housing.

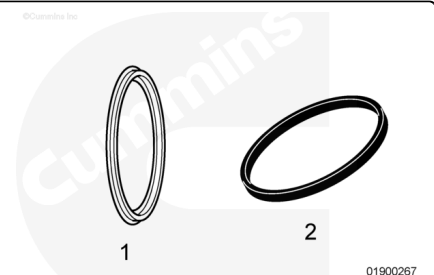
Note : Make sure the seal is positioned square with the crankshaft.

Remove the seal pilot.



Each new lip style crankshaft seal comes with a disposable seal driver.

- Metal Installation Tool (1): This tool should be used for front gear train engines with a rear seal carrier. The installation tool will install the seal to the proper depth in the rear seal carrier bore of ~8 mm [~0.315 in]. (Note: this



is different than the plant installation depth of ~4 mm [~0.157 in])

- Plastic Installation Tool (2): This tool should be used for rear gear train engines. The small chamfer on the inner diameter **must** be used to contact the rear crankshaft seal. The installation tool will make sure the seal is **not** installed deeper than 2 mm [0.079 in] into the flywheel housing bore.

Note : For rear gear train engines, the rear crankshaft seals for the wet flywheel housing applications do **not** come with a disposable seal driver. Service tool, Part Number 3824078, **must** be used to install the rear crankshaft seal for the wet flywheel housing. This is the same service tool used for installing the rear crankshaft seal and wear sleeve assembly. Reference Procedure 001-067 in Section 1. For front gear train engines, this service tool can be used in place of the disposable seal driver that comes with the crankshaft seal.

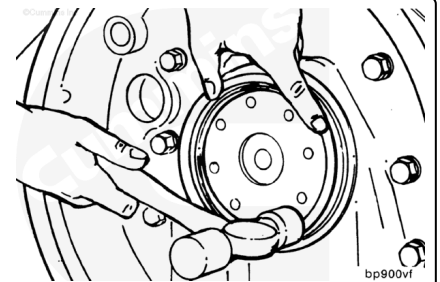
Note : It may be necessary to lightly tap the rear crankshaft seal with a plastic hammer without the disposable seal driver to help get the seal started.

Use the appropriate disposable seal driver that comes with each new rear crankshaft seal to install to the correct depth in the housing or carrier.

Use a plastic hammer to drive the seal into the housing until the alignment tool stops against the housing.

Hit the tool at 12, 3, 6, and 9 o'clock positions to drive the seal evenly and to prevent bending the seal carrier.

After the lip seal installation, the repair can continue. Be careful to **not** bump the part or operate the engine within the first two hours after the seal installation. This allows time for the soapy water or P-80, Part Number 3824878, that was used on the outer diameter of the seal to dry and set.



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

** WARNING **

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the flywheel, if removed. Refer to Procedure 016-005 in Section 16.
(/qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html)
- Install the flexplate, if removed. Refer to Procedure 016-004 in Section 16.
(/qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html)
- Install the transmission and all related components, if equipped. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Start and operate the engine and check for leaks.